
Radar acadêmico semanal – 01/09/2026

1 mensagem

Fabrício Tonetto Londero <fabriciotlondero@hotmail.com>

1 de setembro de 2026 às 09:47

Para: "fabriciotonettolondero@gmail.com" <fabriciotonettolondero@gmail.com>

Radar acadêmico semanal

Artigos publicados entre 22/08/2026 e 01/09/2026.

Jogos educacionais, serious games e aprendizagem baseada em jogos

INTEGRATION OF GAME-BASED LEARNING IN TEACHING MATHEMATICS 7

Autores: Mary Joy B. Etorma**Periódico:** World Journal of Advanced Research and Reviews | **Publicação:** 28/08/2026 |  Acesso aberto**Assunto:** Educational Games and Gamification

Mathematics 7 serves as a crucial foundation for learners transitioning to junior high school, yet many students experience low engagement and difficulty in understanding abstract mathematical concepts. Traditional teaching approaches often fail to address diverse learning needs and motivation levels. In response, game-based learning (GBL) has emerged as a learner-centered instructional strategy that integrates play, challenge, and feedback to enhance understanding and participation. This study examined the integration of game-based learning—both ICT-based and non-ICT-based—in teaching Mathematics 7 and its influence on learner performance and classroom instruction using a quasi-experimental

Defining Game-Based Learning

Autores: Elīna Grāvelsiņa, Linda Daniela**Periódico:** International Journal of Game-Based Learning | **Publicação:** 28/08/2026 |  Acesso aberto**Assunto:** Educational Games and Gamification

Game-based learning (GBL) has emerged as a widely discussed educational approach; however, this term may be referred to by various names, creating challenges for both research and practice. This article examines terminological gaps in the existing discourse on GBL and distinguishes it from related concepts such as gamification, serious games, etc. The focus is on understanding GBL and the related terminology provided. For this reason, a systematic literature analysis of 52 articles was conducted. Key findings contain an overview of the definition of “game-based learning” and its 72 variations. To gain a better understanding, terms were analyzed to answer the following questions: what specifi

Effectiveness of digital game-based learning in elementary education: A systematic literature review

Autores: Fauzia Hasanah, Sandi Budi Iriawan, Effy Mulyasari, Zsazsa Puspa Alivia**Periódico:** Primary Jurnal Pendidikan Guru Sekolah Dasar | **Publicação:** 30/08/2026 |  Acesso aberto**Assunto:** Educational Games and Gamification

This study aims to systematically synthesize empirical evidence on the effectiveness of Digital Game-Based Learning (DGBL) in elementary education, focusing on both academic achievement and affective learning outcomes. A systematic literature review was conducted following the PRISMA protocol to ensure methodological rigor and transparency. Peer-reviewed empirical articles published between 2021 and 2025 were retrieved from the Scopus database using pre-defined inclusion and exclusion criteria. The selection process involved identification, screening, eligibility assessment, and final inclusion stages, resulting in 20 eligible studies. Data were analyzed using thematic synthesis to identify

Effect of Game-Based Learning on Students' Retention Of Knowledge Among Undergraduate Medical Students

Autores: Fahima Arefin, Shahin Akhter, Burhan Uddin, Debjane Barua

Periódico: Journal of Chittagong Medical College Teachers Association | **Publicação:** 25/08/2026 | ● Acesso aberto

Assunto: Educational Games and Gamification

Background: Successful medical education is essential for medical students' professional proficiency. Interactive teaching strategy helps the medical undergraduates better understanding in dynamic way. One of the most important goal of this teaching method is to create an environment which promotes students' motivation and interest and thus facilitates learning. The aim of this study was to assess medical students' retention of knowledge using gamebased learning method. Materials and methods: This experimental study was conducted in the Department of Physiology, Chittagong Medical College and Institute of Applied Health Science, Chattogram. Total 126 apparently healthy 1st year medical stude

Beyond The Game: A Phenomenological Study On the Lived Experiences of Game-Based Learning in Physical Education and Health Classes

Autores: Liane Kristal L. Pabustan, Samboy G. Nabor

Periódico: INTERNATIONAL JOURNAL OF MULTIDISCIPLINARY RESEARCH AND ANALYSIS | **Publicação:** 25/08/2026 | ● Acesso aberto

Assunto: Physical Education and Pedagogy

This study explored the lived experiences of game-based learning approaches in Physical Education and Health classes as basis for developing integrative lesson exemplars for Grade 7 learners. It aimed to describe the learners' experiences with game-based learning, derive meanings from these experiences, identify the major themes that emerged from the coded responses, and use the findings as basis for a framework and sample lesson exemplars. The study employed a qualitative research design using semi-structured interviews with eight Grade 7 participants who had been exposed to game-based and game-based activities in PEH classes. Interview responses were transcribed, coded, and analyzed through

Gamificação e aprendizagem gameful

Gamificação no Ensino à luz da Teoria da Autodeterminação: uma análise textual das implicações para motivação, engajamento e aprendizagem

Autores: Leandro Blass, Eduarda Da Silva Gonçalves

Periódico: Educação Matemática em Revista - RS | **Publicação:** 23/08/2026 | ● Acesso aberto

Assunto: Motivation and Self-Concept in Sports

This study analyzed, in light of Self-Determination Theory, the contributions of gamification to motivation, engagement, and learning in education. It is a qualitative literature review based on studies published between 2015 and 2025 and retrieved from Google Scholar and the Brazilian Digital Library of Theses and Dissertations. After applying the selection criteria, the corpus comprised four articles and three academic works, which were subjected to textual analysis using

IRaMuTeQ software through Descending Hierarchical Classification. The results indicate that gamification, when pedagogically planned, promotes autonomy, perceived competence, relatedness, persistence in the face of challenge

The Impact of Gamified Learning on Student Achievement: A Narrative Literature Review

Autores: Dennis Jay Calang, Mary Joy Abadinas, Vilma Bon

Periódico: JPAIR Institutional Research | **Publicação:** 23/08/2026 |  Acesso aberto

Assunto: Educational Games and Gamification

This narrative literature review explores the statistically significant relationship between gamified learning, student engagement, and academic achievement. Drawing from quasi-experimental studies, meta-analyses, and systematic reviews, the paper highlights how game-based elements—such as points, badges, leaderboards, and immediate feedback—enhance motivation and foster higher-order thinking skills. The review also examines the differential impact of gamification across achievement levels and the moderating role of socio-demographic variables such as age, sex, parental education, and school type. Findings suggest that while gamification is broadly effective, its success depends on thoughtful

Gamificação na educação inclusiva: revisão da literatura

Autores: Leandro Cosme Martins da Silva, Claudinei Martins Guimarães

Periódico: Cadernos Cajuína | **Publicação:** 25/08/2026 |  Acesso aberto

Assunto: Health Education and Validation

A gamificação tem se destacado no campo educacional como uma estratégia metodológica capaz de incorporar elementos de jogos aos processos de ensino e aprendizagem, favorecendo o engajamento, a motivação e participação dos estudantes. No contexto da educação inclusiva, essa abordagem apresenta potencial para contribuir com práticas pedagógicas mais acessíveis, interativas e adequadas às diferentes necessidades educacionais. Nesse sentido, o presente estudo teve como objetivo analisar as evidências científicas acerca da utilização da gamificação em contextos da educação inclusiva, identificando as estratégias empregadas, seus fundamentos teóricos, potencialidades, desafios e contribuições para

A importância da gameficação no ensino de Química

Autores: Paulo Henrique Almeida da Hora

Periódico: Revista Interseção | **Publicação:** 30/08/2026 |  Acesso aberto

Assunto: Science and Education Research

RESUMO: O ensino de Química, ao longo dos anos, tem enfrentado desafios relacionados à abstração dos conteúdos e à dificuldade dos estudantes em compreender sua aplicação prática no cotidiano. Essa problemática reflete um modelo de ensino tradicional, centrado na memorização e na exposição teórica, que frequentemente distancia o aluno do papel ativo no processo de aprendizagem (Kalogiannakis; Papadakis; Zourmpakis, 2021). Nesse cenário, a gamificação surge como uma estratégia pedagógica inovadora capaz de romper barreiras cognitivas e emocionais, tornando o aprendizado mais envolvente, dinâmico e significativo. Este artigo tem como objetivo discutir a importância da gamificação no ensino de

Letramento em jogos e formação de professores

Implementation of problem-based learning using domino cards to increase learning activity and achievement mathematics

Autores: Heri Siswanto, Joko Sulianto, Ngatmini

Periódico: Indonesian Journal of Educational Development (IJED) | **Publicação:** 26/08/2026 |  Acesso aberto

Assunto: Mathematics Education and Pedagogy

Limited classroom participation and low mathematics achievement remain challenges in early elementary education. This study examined (1) the effect of Problem-Based Learning (PBL) using domino cards on mathematics learning activity, (2) its effect on mathematics achievement, and (3) its simultaneous effect on both outcomes. The population comprised 40 first-grade students at SDN 2 Plosorejo, Blora Regency, Indonesia. Using purposive sampling, 20 students participated in each group. A quantitative quasi-experimental method with a Nonequivalent Control Group Pretest–Posttest Design was employed. Learning activity was measured through classroom observation, while achievement was assessed using

Jogos digitais na formação de professores

Teaching videos in initial teacher education: a qualitative study of pre-service teachers' reflections

Autores: Pascal Hohaus

Periódico: Frontiers in Education | **Publicação:** 31/08/2026 |  Acesso aberto

Assunto: Literacy, Media, and Education

Introduction Teaching videos, defined here as recordings of authentic classroom practice in schools, have become an important resource in initial teacher education. This study examines how pre-service teachers of English as a foreign language (EFL) reflect on the use of teaching videos during a university seminar in EFL pedagogy and linguistics. Methods The sample comprised 28 pre-service teachers. Participants viewed selected excerpts from a publicly accessible video documenting an upper-secondary lesson on countering disinformation as part of critical media education. They subsequently took part in a guided discussion. Their written reflections were analysed using reflexive thematic analysis

Defining Game-Based Learning

Autores: Elīna Grāvelsiņa, Linda Daniela

Periódico: International Journal of Game-Based Learning | **Publicação:** 28/08/2026 |  Acesso aberto

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Game-based learning (GBL) has emerged as a widely discussed educational approach; however, this term may be referred to by various names, creating challenges for both research and practice. This article examines terminological gaps in the existing discourse on GBL and distinguishes it from related concepts such as gamification, serious games, etc. The focus is on understanding GBL and the related terminology provided. For this reason, a systematic literature analysis of 52 articles was conducted. Key findings contain an overview of the definition of “game-based learning” and its 72 variations. To gain a better understanding, terms were analyzed to answer the following questions: what specifi

Curriculum Literacy in the Digital Age: Enhancing Pre-Service Teachers' Competence through Digital Literacy

Autores: Serkan Aslan, Üzeyir SÜĞÜMLÜ

Periódico: Participatory Educational Research | **Publicação:** 30/08/2026 |  Acesso aberto

Assunto: Digital literacy in education

Curriculum literacy is a core professional competence that enables pre-service teachers to interpret curricula accurately and translate curricular objectives into effective instructional practices. In digitally mediated learning environments, this competence may also support the purposeful use of digital technologies. This study examined the mediating role of digital literacy in the relationship between pre-service teachers' curriculum literacy and their competence in designing instruction

in digital environments. Using a relational survey design, data were collected from 314 fourth-year pre-service teachers enrolled at two state universities in Türkiye. The Curriculum Literacy Scale, Digita

The Effects of Educational Digital Game Design by Pre-Service Foreign Language Teachers on Their Creative Thinking Dispositions and Attitudes: A Document Analysis Authors Burak Demir

Autores: Burak Demir, Çağla Deniz Özçelik, Yücehan Yücesoy, Behçet Öznacar

Periódico: European Journal of Education and Counselling | **Publicação:** 28/08/2026

Assunto: Educational Games and Gamification

The aim of this study is to examine and thematically evaluate findings reported in the national and international literature regarding the effects of educational digital game design processes on foreign language pre-service teachers' creative thinking dispositions and their attitudes toward digital games and game design, and to propose a conceptual framework based on the findings. The study was conducted using document analysis, a qualitative research method. Studies published between 2010 and 2025 were searched in Web of Science, Scopus, ERIC, DergiPark, and Google Scholar in accordance with predetermined inclusion and exclusion criteria. Of the 96 records identified in the initial search,

Empowering preservice teachers' pck development through resource folio pck development tools: Cores and pap-ers

Autores: Wiputra Cendana, Sean Patrick Kang, Dea Deviani Pradhipta

Periódico: Primary Jurnal Pendidikan Guru Sekolah Dasar | **Publicação:** 30/08/2026 |  Acesso aberto

Assunto: Innovative Teaching and Learning Methods

Effective teaching entails PCK, enabling teachers to integrate different elements of instructional design in dynamic ways. The tacit nature of PCK poses challenges for novice teachers in understanding the reasons behind certain pedagogical approaches throughout the teaching-learning process. Resource Folio, as one of the PCK development tools that combines Content Representations (CoRes) and Pedagogical and Professional-experience Repertoires (PaP-eRs), is expected to be the solution for capturing teachers' implicit pedagogical reasoning embedded in instructional design. This study investigates the use of the Resource Folio in developing and refining topic-specific PCK. 27 4th- to 6th-grade

Jogos digitais como REA e práticas educacionais abertas

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Artificial Intelligence for Enhancing English Learning among Technical College Students in Derna

Autores: Soad Issa

Periódico: AlQalam Journal of Medical and Applied Sciences | **Publicação:** 28/08/2026 |  Acesso aberto

Assunto: Technology-Enhanced Education Studies

This study explored the role of Artificial Intelligence (AI) in enhancing English language learning among students at the Faculty of Technical Sciences in Derna. With the rapid advancement of AI technologies, educational practices incorporated intelligent tools to improve learning outcomes and student engagement. The research adopted a descriptive-analytical approach, utilizing a structured questionnaire to collect data from a sample of 49 students (males and females) at the Faculty of Technical Sciences in Derna. The study examined students' perceptions of using AI-powered applications, such as Google Translate and Duolingo, in developing language skills, including vocabulary, pronunciation

Design e desenvolvimento de jogos educacionais

Reorienting Pedagogy in Secondary Mathematics Education and Evaluating the Effectiveness of Digital Resources under the NEP 2020 Framework

Autores: Saurabh Kapoor, Ashok Kumar Sahoo

Periódico: Journal of Graphic Era University | **Publicação:** 24/08/2026 |  Acesso aberto

Assunto: Open Education and E-Learning

Over the past four years, the Indian education system has increasingly adopted essential 21st-century skills in line with the vision of NEP 2020. Among these, the integration of digital resources has significantly transformed traditional classroom teaching. In mathematics education, digital resources create new learning opportunities and positively influence students' understanding. Therefore, this study examines the effect of digital resources on learning mathematics at the secondary school level. Using survey data collected from different Jawahar Navodaya Vidyalayas (JNVs), the study applies ANOVA and Chi-square tests to analyse the impact of digital resources on students' mathematics learn

ZeroByte: A board game to explore computer science concepts

Autores: João Bernardo, Fernanda Pires, Marcela Pessoa, Rafaela Melo, José Carlos Duarte, Jeniffer Macena de Souza

Periódico: Figshare | **Publicação:** 26/08/2026 |  Acesso aberto

Assunto: Educational Games and Gamification

Background: Games have been increasingly adopted to support learning; however, designing educational games remains challenging due to the need to balance pedagogical and playful elements. Purpose: This paper presents the ideation, development, and evaluation of ZeroByte, an educational board game designed to support the learning and review of introductory Computer Science concepts. The game comprises 180 cards organized into three categories, containing 1,260 clues and 540 luck or setback events. Methods: ZeroByte was developed using an educational game design methodology grounded in Meaningful Learning Theory and the tangential learning approach. Two empirical studies were conducted. First,

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Bibliometric Analysis of Global Research Trends on Learning Basic Programming Through Games

Autores: Shafinaz Mohammad Niyaz Khan, Juzlinda Md. Ghazali, Noor Azli Mohamed Masrop, Roslinda Ramli, Roslina Ibrahim

Periódico: e-Jurnal Penyelidikan dan Inovasi | **Publicação:** 25/08/2026 | ● Acesso aberto

Assunto: Educational Games and Gamification

Games' inclusion into programming education has received increased attention due to their ability to improve student engagement, motivation, and problem-solving skills while learning basic programming concepts. Despite the increasing number of publications in this discipline, a thorough grasp of the research landscape, emerging themes, and publication patterns is still lacking. As a result, this research provides a bibliometric overview of research on games in beginning programming instruction. Bibliographic information was obtained from the Scopus database using specific keyword combinations relating to games, game-based learning, and fundamental programming instruction. The analysis looked

Avaliação, usabilidade e acessibilidade em jogos

Effectiveness of digital game-based learning in elementary education: A systematic literature review

Autores: Fauzia Hasanah, Sandi Budi Iriawan, Effy Mulyasari, Zsazsa Puspa Alivia

Periódico: Primary Jurnal Pendidikan Guru Sekolah Dasar | **Publicação:** 30/08/2026 | ● Acesso aberto

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INTEGRATION OF GAME-BASED LEARNING IN TEACHING MATHEMATICS 7

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Effects of an Artificial Intelligence–Supported Serious Game on Healthy Lifestyle Behaviours: A Quasi-Experimental Study

Autores: Afra Çalık, Osman Yıldırım

Periódico: Süleyman Demirel Üniversitesi Sağlık Bilimleri Dergisi | **Publicação:** 24/08/2026 | ● Acesso aberto

Assunto: Mobile Health and mHealth Applications

Aim: This study aimed to evaluate the effects of an AI-powered serious game on healthy lifestyle behaviours among adults undergoing diabetes risk assessment. **Methods:** A single-group quasi-experimental design was conducted with 60 adults without a diabetes diagnosis (FINDRISK score ≥ 12). The Health-Promoting Lifestyle Profile-II (HPLP-II) was administered at four time points over six months, assessing health responsibility, physical activity, nutrition, spiritual growth, interpersonal relations, stress management, and total score. Perceived behavioural impact of game mechanics was evaluated using a structured assessment form. Data were analysed using repeated-measures ANOVA with Greenhouse-Ge

Inteligência artificial na educação e formação docente

Artificial intelligence in mathematics education: a two-dimensional conceptual framework and meta-review

Autores: Maike Schindler, Mustafa Çevikbaş, Achim J. Lilienthal

Periódico: ZDM | **Publicação:** 24/08/2026 | ● Acesso aberto

Assunto: Intelligent Tutoring Systems and Adaptive Learning

Abstract Artificial Intelligence (AI) is transforming mathematics education just as it is reshaping many other domains of modern life, creating new opportunities that fundamentally change the way students and teachers work. AI has become an increasingly prominent topic in mathematics education, generating a growing body of systematic reviews. However, the review-level literature remains fragmented across different AI paradigms, system types, and methodological approaches. This meta-review synthesizes existing systematic reviews on AI in mathematics education and provides a conceptually oriented interpretation of how AI is shaping mathematics education. We specifically identify and classify t

Collaborative Inference and Learning between Edge SLMs and Cloud LLMs: A Survey of Algorithms, Execution, and Open Challenges

Autores: Senyang Li, Haozhao Wang, Wenchao Xu, Rui Zhang, Song Guo, Jingling Yuan

Periódico: ACM Computing Surveys | **Publicação:** 24/08/2026 | ● Acesso aberto

Assunto: IoT and Edge/Fog Computing

As large language models (LLMs) evolve, deploying them solely in the cloud or compressing them for edge devices is increasingly inadequate due to latency, privacy, cost, and personalization concerns. This survey examines a collaborative paradigm in which cloud-based LLMs and edge-deployed small language models (SLMs) cooperate for both inference and training. We propose a unified taxonomy of edge-cloud collaboration strategies. For inference, we categorize approaches into task assignment, task division, and mixture-based collaboration at both task- and token-levels, encompassing adaptive scheduling, resource-aware offloading, speculative decoding, and modular routing. For training, we review

A Survey on Human-AI Collaboration with Large Foundation Models

Autores: Vanshika Vats, Marzia Binta Nizam, Minghao Liu, Ziyuan Wang, Richard C. Ho, Mohnish Sai Prasad

Periódico: ACM Transactions on Intelligent Systems and Technology | **Publicação:** 22/08/2026 |  Acesso aberto

Assunto: Scientific Computing and Data Management

As the capabilities of artificial intelligence (AI) continue to expand rapidly, Human-AI (HAI) Collaboration, combining human intellect and AI systems, has become pivotal for advancing problem-solving and decision-making processes. The advent of Large Foundation Models (LFMs) has greatly expanded its potential, offering unprecedented capabilities by leveraging vast amounts of data to understand and predict complex patterns. At the same time, realizing this potential responsibly requires addressing persistent challenges related to safety, fairness, and control. This paper reviews the crucial integration of LFMs with HAI, highlighting both opportunities and risks. We structure our analysis as follows:

De novo design of RNA pseudoknots with deep learning

Autores: Jill Townley, Wipapat Kladwang, David Baker, Hamish M Blair, Christian Choe, Gina El Nesr

Periódico: Science | **Publicação:** 27/08/2026 |  Acesso aberto

Assunto: RNA and protein synthesis mechanisms

RNA design has been hindered by the limited accuracy of 3D structure prediction. Here, we show that intricate RNA structures can be generated with current deep learning tools through accurate de novo design of pseudoknot secondary structures. In an Eterna competition involving 57 pseudoknots, generative AI methods matched experienced human designers in solving most blind challenges, evaluated by single-nucleotide-resolution chemical mapping, compensatory mutagenesis, and cryogenic electron microscopy. AI-generated molecules with accurate secondary structures formed well-ordered 3D folds stabilized by noncanonical tertiary interactions not modeled during design. Success was guided by an RNet

Generative AI use before medical visits: disclosure-item responses, trust, and care-seeking behaviors in a cross-sectional social-media survey in Poland

Autores: Simona Wójcik, Anna Rulkiewicz, Justyna Domienik-Karłowicz

Periódico: Frontiers in Digital Health | **Publicação:** 27/08/2026 |  Acesso aberto

Assunto: Artificial Intelligence in Healthcare and Education

Background Patient-facing generative artificial intelligence (AI) tools such as ChatGPT are increasingly used for health information seeking, yet their role in patient–physician communication remains insufficiently characterized. Using the Network Episode Model as the primary framework, this study examined self-reported pre-consultation AI use, whether respondents reported disclosing that use to a physician, and self-reported AI-associated health-behavior change in a convenience sample of Polish adults recruited via social media. Findings are exploratory. Methods Cross-sectional online survey, anonymous with respect to the investigators (N = 1,044), administered in Polish via Google Forms with

Inteligência artificial aplicada a jogos educacionais

Effects of an Artificial Intelligence–Supported Serious Game on Healthy Lifestyle Behaviours: A Quasi-Experimental Study

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Artificial Intelligence: A Transformative Solution to Educational Accessibility Issues for Learners with Special Needs

Autores: Ajibade Isaac Olufisayo

Periódico: Nusantara Journal of Behavioral and Social Sciences | **Publicação:** 30/08/2026 |  Acesso aberto

Assunto: Assistive Technology in Communication and Mobility

Artificial Intelligence (AI) is creating new opportunities to improve educational accessibility for learners with special needs and disabilities. However, barriers such as inaccessible learning materials, communication difficulties, inflexible instruction, inadequate assistive technologies, limited teacher competence, poor infrastructure, and unequal technology access continue to restrict participation. This paper examines AI as a transformative solution to these challenges, focusing on learners with visual, hearing, intellectual, communication, learning, physical, autism, multiple, and emotional or behavioural disabilities. It discusses applications including computer vision, speech recogni

Generative Artificial Intelligence and Human Artificial Intelligence Interaction in Immersive Environments: Research and Development Directions

Autores: Thu Nguyen, Simeon Simoff, Zhonglin Qu, Rani Adam, Daniel Catchpoole, Paul J. Kennedy

Periódico: International Journal of Human-Computer Interaction | **Publicação:** 26/08/2026 |  Acesso aberto

Assunto: Virtual Reality Applications and Impacts

Generative artificial intelligence (GenAI) in immersive environments (IEs) is rapidly reshaping the ways humans interact with digital content, raising a range of cognitive, social, and ergonomic concerns in interactive computing. Such transformation calls for systematic reviews of the integration of GenAI and Human–AI (HAI) interaction in IEs, as the work

in both areas remains limited. This systematic survey addresses the gap, focusing on GenAI and HAI interaction in IEs. The paper adopts an HAI interaction framework structured around user intent, AI processing, and iterative interaction, with embedded analysis through experiential, educational, and epistemic lenses. This framework clarifies

C-V2X, V2I2P e usuários vulneráveis da via

Evaluating the role of speed reduction on the severity of vehicle-pedestrian crashes: a case study in Palermo, Italy

Autores: Muhammad Yasir, David Logan, Marco Migliore, Gabriele D'Orso, Amjad Pervez, Sjaan Koppel

Periódico: Frontiers in Future Transportation | **Publicação:** 24/08/2026 |  Acesso aberto

Assunto: Traffic and Road Safety

Introduction This study examined the influence of real-world vehicle speeds on injury outcomes in police-reported vehicle-to-pedestrian crashes in Palermo, Italy. Specifically, the study quantified the relationship between percentile travel speeds and the risk of fatal and serious injuries (FSI) for pedestrians and assessed the potential effects of small reductions in travel speed using counterfactual scenarios. **Methods** Pedestrian crash locations from police-reported vehicle-to-pedestrian crashes in Palermo, Italy, were linked with real-world travel speed data obtained from TomTom Move. Percentile travel speeds (15th, 50th, and 85th) were used as proxies for impact speed to estimate FSI risk

Who benefits? Australian public perceptions of safety and justice in connected and automated vehicles (CAVs)

Autores: Laura Martinez-Buelvas, Andry Rakotonirainy, Deanna Grant-Smith, Óscar Oviedo-Trespalacios

Periódico: AI & Society | **Publicação:** 28/08/2026 |  Acesso aberto

Assunto: Human-Automation Interaction and Safety

Abstract The integration of connected and automated vehicles (CAVs) into everyday mobility systems raises profound ethical and societal challenges, particularly concerning safety, justice, and the treatment of vulnerable road users (VRUs). This study examines Australian public perceptions of the fairness, safety, and legitimacy of CAV deployment, investigating how exposure to information related to ethics and justice influences acceptance and evaluative perceptions related to trust (including perceived safety, fairness, and institutional confidence). An online survey of 384 respondents (aged 18–82) employed a before-and-after design, in which respondents evaluated CAVs before and after engag

Ensuring information integrity in V2X cooperative perception systems for multi-agent autonomous vehicles

Autores: И.И. Мыськив, D. A. Zakoldaev, Alexander Menshchikov

Periódico: Scientific and technical journal of information technologies mechanics and optics | **Publicação:** 27/08/2026 |  Acesso aberto

Assunto: Adversarial Robustness in Machine Learning

Cooperative perception based on Vehicle-to-Everything (V2X) technologies is effective methodology that allows the use of various data sources, such as sensors, cameras, and sensors, enabling unmanned vehicles to exchange data to overcome the limitations of individual perception. However, the integration of multi-agent data systems creates additional vector attack in cooperative perception and information integrity protection. A quantitative metric of information integrity $I_{\mathcal{A}}(t, k)$, is introduced, based on pairwise consistency of agent observations using Mahalanobis distance. A threat model is developed that classifies types of attacks in V2X systems. A model has been developed that includ

Optimising Urban Traffic Flow through Green-Wave Signal Coordination: Effects on Delays, Fuel Consumption and Emissions in Baku

Autores: Mirhamid Baghirov, Heybetulla Mabud Əhmədov

Periódico: Communications - Scientific letters of the University of Zilina | **Publicação:** 25/08/2026 |  Acesso aberto

Assunto: Vehicle emissions and performance

In this study it was investigated how the traffic optimisation can reduce the environmental impact of urban transport. Traffic flows were modelled on a key urban road section, and traffic-light phases were optimised and synchronised to create smoother movement and reduce intersection delays. Data on traffic volume, delay time, fuel consumption and pollutant emissions were collected before and after the implementation. The results showed a 20-25% reduction in intersection delays, accompanied by an average of 15% decrease in CO2 emissions. Fuel consumption fell by 10% for cars and 15% for trucks. Traffic-light optimisation also improved the flow distribution and increased average speed. Enviro

Engineering proactive risk fields in mixed-autonomy traffic via anticipatory swarm intelligence: a digital twin perspective

Autores: Qihao Liu, Zhichun Li, Daiquan Xiao, Xuecai Xu, Changxi Ma

Periódico: Scientific Reports | **Publicação:** 29/08/2026 |  Acesso aberto

Assunto: Digital Transformation in Industry

Ensuring safety and resilience in mixed-autonomy traffic systems presents a critical engineering challenge due to the stochastic nature of human-driven vehicles (HDVs) and complex interactions with connected autonomous vehicles (CAVs). Traditional safety models predominantly rely on reactive post-conflict assessment, lacking the capability for proactive intervention. To address this gap, this study proposes an Anticipatory Risk Field Dynamics (ARFD) framework and develops an Anticipatory Swarm Intelligence (ASI) algorithm for proactive risk engineering. Methodologically, the ARFD framework establishes a novel closed-loop formulation of risk evolution governed by partial differential equation

Gêmeos digitais, CARLA e Vehicle-in-the-Loop

Intelligent Transportation Systems: A Review of Integration of Digital Twin and Machine Learning Control

Autores: Thit Tun, Hakilo Sabit

Periódico: IoT | **Publicação:** 27/08/2026 |  Acesso aberto

Assunto: Traffic control and management

This paper presents a comprehensive review of the integration of Intelligent Transportation Systems (ITS) and Digital Twin (DT) technologies for intelligent traffic management. It examines the role of key enabling technologies, including the Internet of Things (IoT), machine learning (ML), deep learning (DL), reinforcement learning (RL), Graph Neural Networks (GNNs), vehicle-to-everything (V2X) communication, and edge computing. These technologies support real-time traffic monitoring, traffic prediction, and adaptive control in ITS. The review synthesizes recent research on conventional traffic control methods, optimization-based approaches, learning-based techniques, and DT-enabled traffic

FKF Transform Based Multi-Scale Analysis and Digital Twin-Enabled Supply-Chain Resilience

Autores: Sharma V A, Shinde S M, Harle S M

Periódico: International Journal of Creative and Open Research in Engineering and Management | **Publicação:** 25/08/2026 |  Acesso aberto

Assunto: Digital Transformation in Industry

The convergence of AI, Blockchain, IoT, Digital Twins, quantum computing, and FKF spectral methods constitutes a decisive research direction for next-generation logistics, especially when examined through the lens of Smart Mobility and Intelligent Transportation Systems [9]. Future work must prioritize real-time fusion of connected-vehicle and smart-infrastructure data streams, construction of scalable multi-resolution Digital Twin environments that span both supply chains and urban mobility networks, development of trustworthy AI models for predictive and prescriptive control, realization of practical hybrid quantum–classical optimizers for the combinatorial problems that dominate intelligence

Engineering proactive risk fields in mixed-autonomy traffic via anticipatory swarm intelligence: a digital twin perspective

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Assunto: Digital Transformation in Industry

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Efficient Spatiotemporal Traffic Forecasting for Smart Cities Using DCRNN-Lite in a Digital Twin Environment

Autores: Hussain Ahmad Awad

Periódico: International Journal Research on Metaverse. | **Publicação:** 22/08/2026 | ● Acesso aberto

Assunto: Traffic Prediction and Management Techniques

Urban traffic congestion remains a critical challenge for smart city development, particularly in the context of real-time monitoring and prediction for intelligent transportation systems. To address this issue, this study proposes a Digital Twin-based urban traffic prediction framework using a lightweight Diffusion Convolutional Recurrent Neural Network (DCRNN-Lite). The proposed model integrates spatial dependencies among road segments through diffusion convolution and temporal traffic dynamics through recurrent modeling, enabling effective spatiotemporal traffic forecasting with reduced computational complexity. Experiments were conducted on the real-world METR-LA dataset, consisting of t

An FKF Spectral Framework for Digital Twin Enabled Prediction

Autores: Shinde SM, Harle SM

Periódico: International Journal of Creative and Open Research in Engineering and Management | **Publicação:** 25/08/2026 | ● Acesso aberto

Assunto: Digital Transformation in Industry

This study introduces an FKF-enabled intelligent supply-chain framework that integrates Artificial Intelligence (AI), Blockchain, Internet of Things (IoT), Digital Twins, and quantum optimization into a unified architecture. The FKF transform provides a mathematical spectral representation of supply-chain signals, enabling the identification of temporal shifts, modulation effects, multiscale patterns, demand fluctuations, and lead-time dynamics. These spectral features can be supplied to AI and machine-learning models to improve forecasting, anomaly detection, disruption prediction, and resilience assessment. IoT devices continuously provide real-time operational data from transportation, in

ADAS, AEB e detecção de pedestres

Investigating the role of Advanced Driver Assistance Systems (ADAS) in reducing sideswipe collisions in rural Ohio

Autores: Abdul S. Ngereza, Boniphace Kutela, Ibrahim Ibrahim, Panick Kalambay, Ntemi Masanja, Emmanuel Kidando

Periódico: Journal of Safety Research | **Publicação:** 27/08/2026 | ● Acesso aberto

Assunto: Traffic and Road Safety

Introduction: Rural roadways continue to experience disproportionately high fatality rates per vehicle-mile traveled compared with urban facilities, underscoring persistent safety challenges that demand targeted interventions. Among the most prevalent crash types on rural roads are sideswipe collisions, frequently attributed to driver error, limited roadway delineation, and reduced situational awareness. Advanced Driver Assistance Systems (ADAS), including Blind Spot Warning (BSW), Lane Departure Warning (LDW), and Lane Keeping Assistance (LKA), offer potential countermeasures by enhancing driver perception and lateral control; however, existing effectiveness evidence is predominantly freque

STG: Structured Topology of Gridpoints for Occluded Pedestrian Detection

Autores: Tian Qiu, Jifeng Shen, Xin Zuo

Periódico: Sensors | **Publicação:** 30/08/2026 | ● Acesso aberto

Assunto: Advanced Neural Network Applications

Pedestrian detection in crowds is a challenging problem in computer vision. Existing occlusion-handling methods heavily rely on expensive visible-box annotations to locate visible body parts, posing severe limitations in label acquisition cost and open-world generalization. To break through this limitation, we propose a novel Structured Topology of Gridpoints (STG) framework. Operating strictly under standard full-box annotations without any extra visibility supervision, STG aims to achieve implicit, fine-grained local semantic compensation. Specifically, we formulate a coarse-to-fine reasoning paradigm consisting of three interactive stages. To mitigate the high spatial complexity and elimi

A unified lightweight ADAS architecture integrating traffic sign recognition, forward collision warning, and bimodal distillation for autonomous perception

Autores: M. Maddan, N. Sankkarshana, M.Gayathri, Rahul S.G., T. M. Amirthalakshmi

Periódico: Frontiers in Future Transportation | **Publicação:** 24/08/2026 | ● Acesso aberto

Assunto: Advanced Neural Network Applications

In an autonomous driving vehicle, the onboard perception system is essential for understanding a dynamic, complex road scene in real time, which is fundamental to autonomous driving. Perception tasks, such as object detection, classification of traffic signs, lane detection, etc., have seen significant progress in autonomous vehicles. One of the major practical challenges is designing an integrated, computationally efficient system capable of supporting multiple perception tasks without compromising detection performance or increasing the hardware cost of embedded automotive platforms. The proposed work introduces an integrated module-based Unified Advanced Driver Assistance System (ADAS) ar

Low-Light Pedestrian Detection Toward Nighttime Safety Monitoring in Smart Built Environments: A Frequency-Aware RGB–Infrared Fusion Approach

Autores: Chao Zhang, Xingkun Li, Xiangyang Cao

Periódico: Buildings | **Publicação:** 28/08/2026 | ● Acesso aberto

Assunto: Video Surveillance and Tracking Methods

Reliable pedestrian perception under low illumination is important for nighttime monitoring in smart built environments. However, visible-light detectors often lose texture and edge information, whereas conventional RGB–infrared fusion may introduce cross-modal noise and discard discriminative cues during scale conversion. This study proposes Multimodal Wavelet–Spectral DETR (MWSD), a frequency-aware RGB–infrared detection framework. MWSD employs a dual-branch

Multimodal Fusion Feature Sampling backbone for cross-modal interaction. The Multimodal Frequency-Domain Feature Enhancement (MFFE) module produces input-dependent Fourier modulation within shared detection features, rather than recons

Vehicle-Specific Underbody Collision Prediction Using Vision, Vehicle Dynamics, and Physics-Guided Machine Learning

Autores: Dr. Pankaj Malik, Yash Upadhyay, Swaraj Singh Rajput, Anuj Patidar, Taher Fakhri

Periódico: Zenodo (CERN European Organization for Nuclear Research) | **Publicação:** 25/08/2026 | ● Acesso aberto

Assunto: Automotive and Human Injury Biomechanics

Underbody collision is a vehicle-specific safety problem encountered while traversing potholes, speed breakers, road humps, deep depressions, rocks, and uneven or off-road surfaces. Conventional vision-based advanced driver assistance systems (ADAS) can detect road anomalies, but anomaly detection alone does not determine whether a particular vehicle will experience underbody contact. This paper presents an Intelligent Vehicle-Specific Underbody Collision Prediction (IV-UCP) framework that integrates forward-camera road-obstacle detection, road-geometry estimation, vehicle geometry, vehicle dynamics, and a physics-guided minimum-clearance model. The framework represents road geometry, vehicl

Mobilidade para pessoas com deficiência visual

ACESSIBILIDADE ARQUITETÔNICA E AUTONOMIA: UM LEVANTAMENTO DAS NORMATIVAS PARA DEFICIÊNCIA VISUAL

Autores: Carlos Dutra Guedes Junior, Maria de Lurdes Costa Domingos, Mirian Picinini Méxas

Periódico: Revista Acadêmica Online | **Publicação:** 28/08/2026 | ● Acesso aberto

Assunto: Health, Education, and Aging

A inclusão de pessoas com deficiência nos espaços educacionais brasileiros é resultado de um processo histórico marcado por exclusões e avanços legais relativamente recentes, especialmente a partir da consolidação de políticas públicas que passaram a garantir o direito à educação de forma mais ampla. No entanto, ainda existem lacunas significativas entre o acesso formal e a permanência efetiva desses estudantes nas instituições de ensino. O objetivo geral consiste em sistematizar os requisitos técnicos e legais necessários para garantir a mobilidade e segurança dos estudantes cegos no ambiente universitário. O presente estudo caracteriza-se como uma pesquisa de abordagem qualitativa, de natu

A Perception-Action-Semantics Loop for Enhancing Accessible Mixed Reality Heritage Exploration for Blind and Low Vision Users

Autores: Feng Tian, Gong Yan, 谭子扬, Qiaolian Zhu, Haibo Xu, Yuzhi Li

Periódico: International Journal of Human-Computer Interaction | **Publicação:** 24/08/2026

Assunto: Tactile and Sensory Interactions

Cultural heritage exploration for blind and low vision (BLV) users requires coordinated multimodal support rather than isolated sensory substitution. We developed an accessible mixed reality (MR) system integrating tactile exploration, navigation support, and an LLM-based conversational agent, and conducted a counterbalanced mixed-methods study with 20 BLV participants across three interaction modes. Tactile-grounded semantic interaction was associated with stronger aesthetic engagement, greater perceived autonomy, higher flow, and longer artifact-centered exploration without increasing cognitive load. These findings suggest that interaction coherence, rather than modality accumulation, bett

Ensino de Matemática para alunos com deficiência visual no contexto educacional cearense: uma revisão sistemática de literatura

Autores: Cleilson Silva Alves, Francisco Herbert Lima Vasconcelos, Robéria Vieira Barreto Gomes

Periódico: Revista Cearense de Educação Matemática | **Publicação:** 31/08/2026 | ● Acesso aberto

Assunto: Health, Education, and Aging

This study is situated within the research line Digital Technologies in the Teaching of Science and Mathematics of the Graduate Program in Science and Mathematics Education at the Federal University of Ceará (PPGENCIMA/UFC). It presents a Systematic Literature Review (SLR) addressing the teaching of Mathematics to students with visual impairments in the educational context of the state of Ceará, Brazil. Considering that Mathematics is characterized by a high level of abstraction and the predominance of visual representations, such features may constitute barriers to the teaching and learning process for this population, thus requiring the adoption of accessible pedagogical strategies and inc

Opinions of Visually Impaired University Students on Assistive Technology Use

Autores: İbrahim Bingül, Ayşe Tuba Ceyhun

Periódico: Van Yüzüncü Yıl Üniversitesi Eğitim Fakültesi dergisi/Yüzüncü Yıl Üniversitesi Eğitim Fakültesi dergisi |

Publicação: 26/08/2026 | ● Acesso aberto

Assunto: Assistive Technology in Communication and Mobility

This study explores the perspectives and experiences of university students with visual impairments regarding their use of assistive technology in both academic and daily life. Employing a qualitative phenomenological research design, the study investigates the lived experiences of these students in relation to assistive technology. Ten visually impaired students from various higher education institutions in Turkey participated in the study; they were selected using purposive sampling. Data were collected through semi-structured interviews, focusing on the participants' experiences using assistive technologies for education, daily living, and accessing information. Content analysis was used

Spatial cognition in hospital buildings for different types of visual impairment: a framework based on Grounded Theory

Autores: Yue Wu, Xiao Du, Fangfang Liu, Jian Kang

Periódico: The Journal of Architecture | **Publicação:** 25/08/2026 | ● Acesso aberto

Assunto: Tactile and Sensory Interactions

This study aimed to construct a systematic framework of spatial cognition for visually impaired individuals, elucidating the mechanisms through which spatial information is transformed into navigational behavior. Using a Grounded Theory approach, fifty-six participants were interviewed, leading to the identification of five key domains: effective environmental information, multi-source perception, perceptual information processing, spatial cognition acquisition, and prompting behaviours. These domains collectively describe a hierarchical pathway from environmental perception to behavioral response. We traced how each impairment group engaged cognitive strategies across key stages of the path

Veículos elétricos, som e percepção de pedestres

Analytical study on the exposedness of potential positions for external human-machine interfaces

Autores: Jose Angel Gonzalez-Belmonte, Jaerock Kwon

Periódico: Autonomous Intelligent Systems | **Publicação:** 26/08/2026 | ● Acesso aberto

Assunto: Ergonomics and Musculoskeletal Disorders

As we move towards a future of autonomous vehicles, questions regarding their method of communication have arisen. One of the common concerns is the placement of the signaling used to communicate with pedestrians and road users, but few works have been fully dedicated to the matter. This paper uses a simulation made in the Unity game engine to record the fifteen different vehicles under fifty-seven different scenarios each for the first time, in order to find how often their forward-facing exterior surfaces can be seen by a pedestrian on the sidewalk. Variables include the vehicle type, position, number of vehicles on the road, camera position and direction, as well as its minimum and maximum

Perceptual and Acoustic Implications of Electric Powered Two-wheelers in Urban Environment

Autores: Dimitris Margaritis, Aikaterini Toulou, Evangelos Bekiaris, Haibo Chen

Periódico: Periodica Polytechnica Transportation Engineering | **Publicação:** 31/08/2026 |  Acesso aberto

Assunto: Noise Effects and Management

The increasing adoption of electric Powered Two-Wheelers (e-scooters and e-motorcycles) in urban mobility has introduced a new safety challenge: their low noise emissions. While reduced acoustic pollution is environmentally advantageous, it may impair the auditory detectability of such vehicles by vulnerable road users, such as pedestrians and cyclists. This study aimed to investigate both the noise characteristics and perceptual detectability of electric versus conventional internal combustion engine (ICE) scooters under simulated urban conditions. Noise measurements were taken for two scooters – one electric and one ICE-powered – at three speeds (10, 20, and 30 km/h). Sixteen participants

Who benefits? Australian public perceptions of safety and justice in connected and automated vehicles (CAVs)

Autores: Laura Martinez-Buelvas, Andry Rakotonirainy, Deanna Grant-Smith, Óscar Oviedo-Trespalacios

Periódico: AI & Society | **Publicação:** 28/08/2026 |  Acesso aberto

Assunto: Human-Automation Interaction and Safety

Abstract The integration of connected and automated vehicles (CAVs) into everyday mobility systems raises profound ethical and societal challenges, particularly concerning safety, justice, and the treatment of vulnerable road users (VRUs). This study examines Australian public perceptions of the fairness, safety, and legitimacy of CAV deployment, investigating how exposure to information related to ethics and justice influences acceptance and evaluative perceptions related to trust (including perceived safety, fairness, and institutional confidence). An online survey of 384 respondents (aged 18–82) employed a before-and-after design, in which respondents evaluated CAVs before and after engag

Review on Thermal Management System of Electric Vehicles

Autores: Mr. Akshay B. Akhade

Periódico: International Journal for Research in Applied Science and Engineering Technology | **Publicação:** 25/08/2026 |  Acesso aberto

Assunto: Electric and Hybrid Vehicle Technologies

The rapid growth of electric vehicles (EVs) has transformed the transportation sector by reducing dependence on fossil fuels and minimizing greenhouse gas emissions. However, the reliability, safety, and efficiency of EVs strongly depend on effective thermal management. Electric vehicle subsystems such as lithium-ion batteries, electric motors, power electronic converters, and charging systems generate substantial heat during operation. Excessive temperature rise can lead to reduced efficiency, accelerated degradation, lower driving range, and severe safety concerns including thermal runaway. Therefore, a robust Thermal Management System (TMS) is essential to maintain optimal operating condi

CAN Bus, telemetria e redes automotivas

MOO-IDS: multi-objective optimization-based lightweight intrusion detection system for in-vehicle networks

Autores: Yali Hao, He Bai, A. Siya, Jiaqi Zhang, Xueliang Fu, Honghui Li

Periódico: Scientific Reports | **Publicação:** 30/08/2026 | ● Acesso aberto

Assunto: Vehicular Ad Hoc Networks (VANETs)

With the proliferation of intelligent connected vehicles, the Controller Area Network (CAN) bus, as the backbone of in-vehicle communication, is vulnerable to cyberattacks due to lack of authentication and encryption. Existing Intrusion Detection Systems (IDS) exhibit limitations in addressing data imbalance, complex attack recognition, and resource-constrained deployment. This paper proposes a lightweight intrusion detection system based on multi-objective optimization, termed MOO-IDS. The system integrates a Hybrid Deep Restricted Boltzmann Machine Generator (HyDRBM-Gen) to mitigate attack sample scarcity and class imbalance in vehicular network data. To address targeted ID fabrication and

DESIGN AND IMPLEMENTATION OF A CAN-BASED SMART VEHICLE DIAGNOSIS SYSTEM WITH CYBER THREAT IDENTIFICATION

Autores: Não informados

Periódico: International Research Journal of Modernization in Engineering Technology and Science | **Publicação:** 31/08/2026 | ● Acesso aberto

Assunto: IoT and GPS-based Vehicle Safety Systems

This paper introduces a new design of the intelligent vehicle diagnostic system that has capability to detect the real-time vehicle parameter and faults through the Controller Area Network (CAN) bus. In the newly introduced design, a dual-ECU design has been adopted in which the information from the sensor such as seat position, ambient light, and distance sensors are collected by a Body Control Module (BCM) and encoded into the standard DTC (Diagnostic Trouble Codes) and transmitted over the CAN bus. A diagnosis ECU receives those messages, decodes the message and collects the sensor reading and the presence of fault. Unlike the conventional OBDs solutions that require external scanning tools

Privacy-preserving multiclass intrusion detection in in-vehicle CAN networks using federated learning

Autores: Ahcène Bounceur, Muhammad Ibrahim, Muhammad Saqib, Nisar Ahmed, Sohail Jabbar, Muhammad Munwar Iqbal

Periódico: Scientific Reports | **Publicação:** 26/08/2026 | ● Acesso aberto

Assunto: Vehicular Ad Hoc Networks (VANETs)

The Controller Area Network (CAN) bus serves as the primary communication backbone in modern connected vehicles. However, the lack of built-in encryption and authentication mechanisms makes CAN-based systems vulnerable to a wide range of cyberattacks, including spoofing, denial-of-service (DoS), and replay attacks. These security weaknesses pose significant challenges for the safe operation of intelligent transportation systems. This paper proposes a privacy-preserving intrusion detection system based on federated learning for multiclass classification of CAN bus traffic, using the CICIOT2024 dataset. Two federated deep learning models, namely Fed-DLSTM and a hybrid Fed CNN-BiLSTM, are devel

LiDAR, ROS e navegação autônoma

Research on Low-Speed Unmanned Vehicle Obstacle Avoidance Based on FeiTengPai

Autores: Lang'en Ni

Periódico: Applied and Computational Engineering | **Publicação:** 23/08/2026 | ● Acesso aberto

Assunto: Autonomous Vehicle Technology and Safety

This paper focuses on low-speed unmanned ground vehicles equipped with the Leishen C16 lidar and the FT-Pi embedded platform. Centering on the problems of autonomous navigation and dynamic obstacle avoidance in unknown environments, it systematically expounds the overall scheme design, engineering implementation and real-vehicle test verification of the system. The system takes a 24-dimensional state vector as input — 20 dimensions are lidar ranging data processed by sector partitioning, and the remaining 4 dimensions are the robot's own state information — and outputs continuous speed control commands accordingly. To coordinate the inherent contradictions among approaching the target, avoid

A new method for estimating forest aboveground biomass by integrating sUAV and satellite remote sensing

Autores: Taiyong Ma, Yang Hu, Mukete Beckline, Xiangming Xiao

Periódico: Australian Forestry | **Publicação:** 30/08/2026 | ● Acesso aberto

Assunto: Remote Sensing and LiDAR Applications

Aboveground biomass (AGB) is a fundamental indicator for assessing forest ecosystem productivity, biodiversity, and carbon sequestration levels. In AGB remote sensing estimation, traditional methods face challenges such as the limited number of plots and issues with matching plots to remote sensing pixels. The insufficient number of plots restricts spatial coverage, while difficulties in plot-to-pixel matching constrain the application of remote sensing data. The use of small Unmanned Aerial Vehicle (sUAV) Light Detection and Ranging (LiDAR) in forest AGB estimation provides a new solution for high-precision and efficient forest resource monitoring. sUAV LiDAR enables the acquisition of deta

LiDAR-based navigation system for spiral-driven grain silo robots using enhanced D*lite

Autores: Haoran Ma, Fengmin Guo, Yue Li, Yue Li, Deliang Dong, Fuyou Wang

Periódico: Scientific Reports | **Publicação:** 27/08/2026 | ● Acesso aberto

Assunto: Image and Object Detection Techniques

Autonomous detection in grain silos faces unique challenges due to confined spaces and dynamic obstacles. Conventional path planning methods often fail to simultaneously meet the requirements of navigation efficiency and trajectory smoothness. The study develops a LiDAR-based system, which is characterized as follows: (1) It employs Gmapping with 5 cm resolution grid mapping. (2) It combines a distance-weighted heuristic ($w = 1.2$) and an enhanced D*lite algorithm for point density optimization. (3) It's three-layer architecture for real-time sensor data fusion. The system was tested in both simulated and real silo environments (12 m in diameter and 30 m in height). The outcomes showed that t